PROJECT PROPOSAL / 01

MARGINA

AI Business Performance Intelligence

Find what slows your business. Measure what it costs.

A unified product combining margin intelligence and process intelligence. Connect sales, costs and operational events to investigate profit changes, identify bottlenecks and evaluate improvement scenarios with traceable evidence.

PROPOSAL	DETAIL
Prepared for	Huzaifa | AI engineering portfolio and business product
Initial customer	A distributor with order, cost and order-to-dispatch event records
Experience design	Guided recruiter sandbox + authenticated business workspace
Planning horizon	20 weeks for a bounded functional pilot; enterprise hardening follows
Document status	Implementation proposal, version 1.0 | 28 October 2026

What this deliverable includes

Product requirements, interface mockups, recruiter walkthrough, production workflows, architecture, data contracts, AI/ML design, evaluation, security, release gates and a commercial pilot plan.

Status boundary: This PDF specifies the system to build. Mockups and sample figures are illustrative. It does not represent an implemented, deployed or enterprise-certified application.

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Product thesis and scope

The customer problem

Finance can see margins and operations can see delays, but the records often live in separate systems. MARGINA links both views at order level so teams can investigate whether costly orders also share operational patterns. It supports decisions without claiming that observed correlation proves causation.

USER	JOB TO BE DONE
Finance manager	Explain contribution-margin changes and inspect transaction-level evidence.
Operations manager	Locate long waits, repeated steps and SLA breaches; prioritize investigation.
Analyst	Reconcile inputs, compare cohorts and export reproducible findings.
Administrator	Manage access, source mappings, retention and integration health.
Recruiter	Explore a guided example and inspect engineering evidence in five minutes.

Initial wedge: distribution, order-to-dispatch

One organization, one currency, one agreed cost policy and one workflow. Start with orders, order lines, product costs, shipping charges, returns and timestamped events. Use the same application services for demonstration and business tenants.

Release boundaries

Functional pilot: authenticated uploads, validated financial metrics, event analysis, linked investigations, grounded AI answers, saved scenarios and exports. Later: one customer-selected ERP adapter, event webhooks, delay prediction and enterprise identity integration.

Excluded initially: accounting ledger replacement, automatic supplier payments, payroll monitoring, universal ERP support, autonomous operational changes and causal claims without a suitable study.

A measurable business use case

Illustrative monthly review

A distributor reports higher sales but lower contribution margin. MARGINA reconciles revenue and costs, then segments low-margin orders by branch, delivery mode and approval waiting time. It surfaces a cohort with both elevated shipping expense and long approval waits.

ILLUSTRATIVE INPUT	AMOUNT / DEFINITION
Net revenue	PKR 12,000,000, excluding sales tax and after discounts/returns
Cost of goods sold	PKR 8,400,000 under the approved costing policy
Variable fulfillment costs	PKR 1,200,000; shipping, handling and payment fees
Contribution	PKR 2,400,000 = 12,000,000 - 8,400,000 - 1,200,000
Contribution margin	20.0% = 2,400,000 / 12,000,000

From evidence to action

The manager opens the affected orders, checks shipping invoices and reviews event timelines. A saved investigation records the association and assigns an owner. The owner proposes a limited approval-process change and compares outcomes during a pilot.

Scenario example - assumptions made visible

Assume 100 urgent shipments could use standard delivery, saving PKR 500 each. Gross modeled savings are PKR 50,000. Subtract PKR 10,000 incremental operating cost to show PKR 40,000 estimated net savings. These are illustrative assumptions, not proven recoverable profit. Overlapping scenarios cannot be summed without deduplication.

Business success measures

Track investigation time, accepted findings, verified shipping-cost changes and sustained workflow improvement. The finance owner validates benefit attribution against a baseline; show changes in volume, product mix and seasonality alongside the result.

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Recruiter demo: a complete journey

Provide a public /demo entry point with a seeded, synthetic distributor. No signup is needed to explore this dataset. The demo cannot access customer workspaces or send external messages. Each session receives isolated scenario state and a visible synthetic-data label.

STEP	VISIBLE BEHAVIOR
1. Start guided tour	Explain the business question and data scope; open a three-month comparison.
2. Inspect margin change	Change branch/date filters; open a low-margin product and its source orders.
3. Follow process evidence	Select the same orders in the process map and inspect approval waiting times.
4. Ask a question	Run a live bounded question: Which delayed orders had high shipping costs?
5. Open evidence	Show tool results, metric definitions, source IDs and an association disclaimer.
6. Test a scenario	Edit shipping conversion and cost assumptions; recalculate and save a scenario.
7. Inspect engineering	Open architecture notes, model evaluation results and a trace with sensitive data removed.

Demo acceptance requirements

Filters must update real computed results from seeded records. Drilldowns must resolve actual synthetic source IDs. At least one cold question must run through the real API. Cached answers must be labeled with their generation time; they cannot masquerade as live computation.

Reset restores only the current demo session. Limit sessions and AI requests, cap tool calls and expire abandoned state. If the AI provider is unavailable, retain deterministic analytics and display a clearly labeled sample answer or a retry option.

Ship three scenarios: rising delivery cost, excessive discounting and repeated approval loops. Publish a short video as a fallback, alongside a runnable repository and benchmark report.

Interface concept: recruiter sandbox



Figure 1. Proposed dashboard layout. All values shown are synthetic illustrations. The PDF mockup is static; the implemented interface must support filtering, evidence inspection and scenario recalculation.

ELEMENT	INTERACTION CONTRACT
KPI cards	Display metric definition, date range, source freshness and cost-coverage indicator.
Linked chart	Selecting a branch filters both financial cohorts and process events.
Evidence panel	Show included orders, exclusions, source references and snapshot version.
AI question box	Offer starter questions plus free text; show progress, answer, evidence and limitations.
Engineering view	Expose sanitized tool traces, measured evaluation results and model/prompt versions.

Use keyboard-accessible controls, readable contrast, labeled axes and non-color indicators. Support desktop first, with a stacked layout for smaller screens. Export tables with the active filters and metric version attached.

Business workspace and onboarding

The business entry point requires authentication and verified organization membership. A setup wizard establishes the source data, costing policy, workflow vocabulary and permissions before activating financial reports.

STAGE	FUNCTIONAL REQUIREMENT
Organization setup	Create tenant, currency, reporting timezone, fiscal calendar and business roles.
Data import	Upload CSVs using templates; preview rows and map columns with explicit confirmation.
Validation	Check unique keys, relationships, numeric types, timestamps, event order and totals.
Reconciliation	Show accepted, rejected and duplicate rows plus differences from source control totals.
Policy approval	Finance owner signs off metric definitions, cost allocation and reporting cutoff.
Workspace activation	Publish approved snapshot; clearly display freshness and unresolved limitations.
Operational review	Assign findings, add comments, save scenarios, export a review package and close cases.

Failures must be recoverable

Incomplete costs should suppress contribution margin or display a partial-cost label. Missing events should reduce process coverage, not generate invented timestamps. Failed imports retain row-level errors and can be corrected and retried without duplicating existing transactions.

Freshness and real-time use

CSV imports provide batch freshness. A later signed webhook/API connector can target updates within five minutes after receipt under agreed load. Display source event time, ingestion time and last successful refresh separately. Do not call a batch source real-time.

Business release acceptance

A customer must be able to import its permitted data, reconcile totals, investigate a finding, save an action and export evidence without developer database edits. Customer-specific mappings are configured, versioned and repeatable.

Interface concept: business operations



Figure 2. Business workspace uses the same analytics shell with tenant-specific authorization, data quality and review controls. The example organization name and figures are fictitious.

SCREEN	REQUIRED STATES AND ACTIONS
Overview	Loading, empty, current, stale and partial data; filters apply consistently.
Margin explorer	Drill from branch to order line; show allocated costs and rounding policy.
Process explorer	Show event paths, cohort counts and elapsed-time distributions.
Investigation detail	Evidence, hypothesis, owner, status, comments and change history.
Scenario laboratory	Assumptions, base case, sensitivity range, overlap warnings and saved versions.
Data quality / settings	Mapping versions, quarantined rows, refresh logs, access and retention controls.

An operations manager sees authorized branches; finance can inspect costs; an analyst may investigate without changing policy. Administrative access does not silently grant permission to export all financial records.

Data model and import contracts

ENTITY	MINIMUM FIELDS
Organization / membership	tenant_id, user_id, role, branch scope, status
Order / order line	tenant_id, source_system, order_id, line_id, product_id, customer_id, branch_id, quantity, unit_price, discount, tax, currency, occurred_at
Cost / return	cost_id, order_id or line_id, category, amount, currency, effective_at; return_id, refunded revenue, inventory recovery policy
Process event	event_id, order_id, activity, lifecycle marker if available, occurred_at, ingested_at, source reference
Snapshot / import	snapshot_id, source file hash, mapping_version, accepted/rejected counts, control totals, refresh state
Finding / scenario	finding_id, evidence references, metric_version, owner, status; scenario assumptions and base snapshot
AI run / audit	run_id, tenant_id, authorized scope, tool results, model version, cost, latency, actor, timestamp

Integrity and normalization

Use tenant-scoped composite foreign keys and source-system-qualified identifiers. Store money in decimal types, not binary floating point. Store timestamps in UTC with original timezone metadata; apply the reporting timezone in presentation. Resolve SKU aliases through reviewed mappings.

Define an order-level process case initially. Split deliveries and returns require explicit subcases later. Preserve raw files immutably for the approved retention period; record normalized row provenance. Do not merge ambiguous customers or products automatically.

Import behavior

Support a documented UTF-8 CSV schema and sample files. MVP limit: 20 MB per file and 50,000 rows per import, subject to benchmarking. File hash plus row business keys support idempotency. Corrections create a versioned replacement or adjustment rather than silently changing history.

Metric and process correctness

METRIC	DEFINITION / GUARDRAIL
Net revenue	Gross sales minus discounts and refunded revenue; exclude collected sales tax. Avoid subtracting discounts twice when source sales are already net.
Gross profit	Net revenue minus recognized COGS. Costing and return-recovery policy must be approved.
Contribution	Gross profit minus included variable costs. Exclude costs already counted in COGS. Name included categories explicitly.
Contribution margin	Contribution / net revenue. Show undefined when revenue is zero; flag negative-revenue periods for review.
Order cycle time	Elapsed time from agreed start to terminal event. Report unfinished cases separately.
Stage elapsed time	Time between recorded events. This includes waiting and processing unless start/complete lifecycle events separate them.
Rework / SLA	Count defined repeated transitions and breaches against versioned business-calendar rules.

Financial comparison

Use an exact, ordered waterfall for price, quantity/mix and cost changes; disclose the calculation order because attribution can depend on it. Verify that components reconcile to the total change. Keep currency conversion out of MVP; later use versioned rates and effective dates.

Process analysis

Build a directly-follows graph with transition counts and median/p90 elapsed times. Handle equal timestamps deterministically and flag ambiguous orderings. Exclude corrupt sequences from affected timing metrics and report coverage. Event timestamps alone do not establish staff effort or resource utilization. PM4Py provides a reference implementation path for event-log analysis [1].

Linked intelligence

Join financial and event data through scoped order IDs. Compare similar branches, product mix and order sizes before highlighting associations. Display sample sizes and coverage; avoid presenting a selected cohort as causal proof.

Architecture and service boundaries

LAYER	RESPONSIBILITY / PROPOSED CHOICE
Web client	Next.js + TypeScript; dashboards, import wizard, investigations and recruiter mode.
Application API	FastAPI + Pydantic; authentication, authorization, typed contracts and orchestration.
Ingestion workers	Redis-backed job queue; validate, normalize, reconcile and publish snapshots.
Analytics engine	SQL + Python; versioned financial metrics and process analysis. Start with PostgreSQL; use Parquet/DuckDB for offline research.
AI service	Hosted model behind a provider adapter; structured tool calls and grounded summaries. Embeddings only where semantic matching or documentation search needs them.
Storage	PostgreSQL for records and metadata; private object storage for raw files and exports.
Operations	Docker, CI, structured logs, OpenTelemetry traces, error tracking and budget monitoring.

End-to-end request flow

Upload -> quarantine/validation -> normalized staging -> reconciliation approval -> immutable analytical snapshot -> computed metrics -> authorized tools -> AI explanation -> evidence-linked interface. Only a fully published snapshot becomes visible to normal reports.

Deployment topology

Deploy frontend, API and workers independently. Keep database, queue and object storage private where supported. Separate demo, staging and production credentials and datasets. Use migrations, connection pooling, request limits and environment-based configuration. Kubernetes and microservice proliferation are unnecessary for the first pilot.

Key API contracts

POST /imports; GET /imports/{id}; POST /snapshots/{id}/publish; GET /metrics; GET /process/cohorts; POST /investigations; POST /assistant/runs; POST /scenarios; GET /exports/{id}. Derive tenant scope from the session, validate branch permissions and enforce idempotency for mutations.

AI and machine learning design

Bounded analytical assistant

Interpret question -> resolve scope and metric -> validate structured tool arguments -> execute approved read-only tool -> verify returned numbers -> generate cited explanation. Limit each run to six tool calls, a configurable token budget and a 60-second timeout. Ask clarification for undefined terms such as profit or late.

CAPABILITY	IMPLEMENTATION AND BASELINE
Natural-language analytics	Allowlisted tools: get_margin_breakdown, compare_periods, get_process_cohort, get_order_evidence and simulate_shipping_cost. No unrestricted SQL or code execution.
Field/entity mapping	Exact keys and normalized aliases first; embedding candidates second; human review for ambiguous matches. Evaluate against labeled pairs.
Margin anomalies	Robust cohort thresholds first; compare an unsupervised model only if it improves useful findings on reviewed cases.
Delay prediction	Optional pilot extension: logistic/tree baseline against a majority/rule baseline. Features must exist at prediction time; exclude future dispatch timestamps.
Documentation retrieval	Retrieve authorized metric definitions and source mapping notes. Financial values come from analytical tools, not vector similarity results.

Evidence and uncertainty

Each material numerical claim references a tool result containing snapshot, metric version, filters and source record IDs. Validate numbers before display. Distinguish observed facts, statistical associations, predictions and scenario assumptions. If evidence is missing, return an explicit limitation.

Models and learning

Begin with one model and prompt version. Do not fine-tune until a measured failure pattern justifies it. Version prompts, schemas and tools; record usage without logging unnecessary personal data. Forecast/prediction retraining requires reviewed data and a held-out evaluation, not automatic learning from every user interaction.

Security, privacy and reliability

CONTROL	IMPLEMENTATION REQUIREMENT
Identity and roles	Managed authentication; secure sessions; finance, operations, analyst and administrator scopes. Add enterprise SSO/MFA integration before contracts requiring it.
Tenant isolation	API checks plus database row policies; isolated object prefixes, cache keys, vector filters, jobs and exports. Application role must not bypass row security [2].
Data protection	TLS, encrypted storage, secret manager, short-lived signed downloads and restricted backups. Mask customer identities where not needed.
AI boundaries	Treat imported text as untrusted data. Validate tool arguments and output; test injection, unauthorized retrieval and resource abuse. Use OWASP guidance as a review aid [3].
Retention and deletion	Customer-configured retention; delete records, files, embeddings and derived exports through a tracked job. Document backup expiry behavior.
Audit and response	Append-only application audit events; privileged access logging, incident runbook and restoration exercises. Do not claim immutability without supporting storage controls.

Operational targets - to validate before pilot acceptance

Reference load: 100,000 order lines, 500,000 events and ten concurrent users on a documented deployment. Target cached dashboard p95 under 2 seconds; uncached under 5 seconds; 90% of bounded AI answers within 30 seconds. Measure import throughput and publish supported capacity rather than assuming unlimited scale.

Target 99.5% monthly availability for the pilot only after monitoring exists. Proposed recovery objectives: 24-hour maximum data loss and eight-hour restoration; verify via backup restoration. Queue retries use backoff, dead-letter handling and idempotent jobs. An AI outage must not disable computed dashboards.

Evaluation and release acceptance

All values below are proposed gates, not achieved results. Pin datasets, metric policies, model versions and test hardware. Synthetic truth supports correctness checks; a consented business pilot establishes usefulness.

AREA	RELEASE GATE
Financial correctness	All golden fixtures reconcile to source controls within documented currency-rounding tolerance; exact waterfall reconciliation.
Process correctness	Known synthetic paths, elapsed times, repeated steps and incomplete cases match expected results.
AI quality	At least 120 held-out questions across finance, process, ambiguity and missing data; >=90% correct supported answers, with >=95% numerical claim support.
Access security	All defined cross-tenant/branch access tests denied across API, exports, caches, objects, search and jobs.
Resilience	Duplicate imports create no duplicate business rows; killed worker can resume; stale and partial states are visible.
Predictive ML	Only release if temporal held-out performance beats the stated baseline; report PR-AUC, calibration and recall at a fixed review budget.
Usability	Five role-based tasks completed without developer intervention by pilot users; record friction and time taken.

Evaluation protocol

Reserve final cases before prompt tuning. Use deterministic graders for calculations and tool permissions; have a finance/operations reviewer assess relevance and explanations. A model judge can assist but cannot be the sole correctness check. Re-run stochastic questions over multiple trials and publish variability.

Testing strategy

Unit tests for cost allocation and event timing; integration tests for import-to-report and authorization; browser tests for demo and business journeys; load tests against the stated reference load; adversarial tests for injected instructions and export access. CI blocks releases on critical regressions.

Delivery roadmap and completion gates

PERIOD	DELIVERABLE AND EXIT CRITERION
Weeks 1-2	Interview a finance and operations user; obtain permitted sample data; approve metric definitions and pilot scope. If no event history exists, change pilot source before process work.
Weeks 3-5	Build tenant/auth foundation, schemas, import validation, control-total reconciliation and golden fixtures. Re-import must be idempotent.
Weeks 6-8	Ship margin explorer, process graph and shared cohort filters. Reconcile both modules against reference fixtures.
Weeks 9-11	Add evidence-backed assistant, investigations and deterministic scenario tools. Evaluate against the initial held-out suite.
Weeks 12-13	Complete recruiter sandbox, guided scenarios, engineering view, session reset and outage behavior.
Weeks 14-16	Run an end-to-end business pilot; improve data quality, access controls, imports, recovery and measured performance.
Weeks 17-20	Finalize deployment, load/security checks, documentation, evaluation report and pilot acceptance. Add delay prediction only if core gates already pass.

Effort assumptions

Estimate: 300-400 engineering hours, roughly 15-20 hours/week over 20 weeks. This assumes a bounded source schema, existing web/Python familiarity and access to a domain reviewer. ERP integration, enterprise SSO and complex costing can materially extend the schedule. Replan after the first sample-data import.

Definition of functional completion

Both entry points use the working backend. A business user imports data, reconciles totals, obtains scoped reports, investigates evidence, saves a scenario and exports results. Recovery and authorization gates pass. Recruiters can complete the guided journey without setup. Production deployment is accepted only for the documented scope and capacity.

Commercial pilot and operating plan

First customer and qualification

Seek one distributor with a finance owner, operations owner and exportable order histories. Require permission to process data, a defined reporting policy and a measurable review task. If the customer cannot link costs and events to orders, start with a data-readiness engagement rather than promise combined insights.

Pilot sequence

Agree a baseline review task and success criteria. Load a restricted historical window, reconcile it with finance, conduct weekly investigations and record verified findings. Review action outcomes after an agreed observation period. Move to paid deployment after user acceptance and security/integration review.

COST CATEGORY	PLANNING ALLOWANCE / MONTH (USD)
API, workers and frontend	40-120
Database, storage and backups	30-100
Model API and embeddings	20-100, controlled by usage quotas
Monitoring and email	0-40
Combined pilot allowance	90-360, excluding labor, taxes, ERP access and enterprise identity services

These are budgeting assumptions, not verified provider quotations. Recalculate with selected providers, storage retention and measured requests. A local development environment can use a 16 GB RAM machine and hosted AI; no GPU is required initially.

Commercial packaging hypothesis

Offer a fixed-scope onboarding engagement plus subscription based on data volume, active users and refresh requirements. Price discovery follows customer interviews and measured ROI. Dedicated deployments, premium support and custom integrations require separate scope; do not advertise universal ERP connectivity.

ROI and business acceptance

Estimate benefit as verified staff time saved plus separately validated avoidable costs, minus subscription, implementation and operating effort. Prevent double counting. The business owner signs off reconciled reports, role access, representative investigations, support expectations and recovery procedures.

Risks, portfolio evidence and references

RISK	MITIGATION / DECISION
Missing or inconsistent records	Coverage indicators, quarantined rows, source reconciliation; narrow the scope when evidence is insufficient.
Misleading financial explanations	Versioned metrics, deterministic totals, citations and finance approval.
False causal conclusions	Association labels, comparable cohorts and controlled follow-up studies where feasible.
Too much scope	One industry, workflow and currency; CSV first; one customer-driven integration later.
Model or infrastructure cost	Per-run limits, tenant quotas, cached aggregates, usage monitoring and graceful degradation.
Enterprise sales friction	Pilot evidence, security documentation, deployment choices and honest capability boundaries.

Portfolio and handover package

Deliver a live synthetic demo, three-minute walkthrough, repository with sample data, reproducible deployment instructions, API specification, architecture decisions, metric dictionary, benchmark report, threat model, operator runbook and known limitations. Publish only consented or synthetic examples. Show measured results rather than invented customer savings.

Reference notes

The product scope, interface and targets in this proposal are original design recommendations. These primary references support specific implementation choices; they do not validate demand or certify the product.

[1] PM4Py: event-log formatting and process-analysis examples. github.com/process-intelligence-solutions/pm4py/blob/release/docs/source/getting_started.rst (reviewed 28 October 2026).

[2] PostgreSQL documentation: Row Security Policies. www.postgresql.org/docs/current/ddl-rowsecurity.html (reviewed 28 October 2026).

[3] OWASP GenAI Security Project: LLM application risk guidance. github.com/GenAI-Security-Project/GenAI-LLM-Top10 (reviewed 28 October 2026).

Recommended first action: Secure one representative sample dataset and approve its metric definitions before writing the AI assistant. This tests whether the combined product can produce trustworthy evidence for a real customer.